

Subtract 00:10:00 from timestamps to find place in audio/video recording.

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00:09:55.040 --> 00:10:13.569

Krzysztof Paruch: Okay, so today is gonna be fun, because I've got myself into something I... yeah, we're gonna see, so we're gonna see how clearly, the discussion is going to go. But, like, my main objective for even picking up this research, or this paper, was to understand

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00:10:13.830 --> 00:10:24.489

Krzysztof Paruch: the relationship between, market scoring rules and cost functions. So, like, you remember we're talking about, information aggregation in a wider sense, prediction markets.

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00:10:24.490 --> 00:10:36.240

Krzysztof Paruch: in a narrower sense, kind of like the intricacies of those markets, like, how can you define and design them? And, last time when we spoke, two or three weeks ago.

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00:10:36.400 --> 00:10:52.089

Krzysztof Paruch: We basically said, okay, there are continuous markets that you can implement, similar to an AMM, that would have the nice properties that you would be looking for. And, my challenge here, or, like, kind of the way how I'm

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00:10:52.160 --> 00:10:54.739

Krzysztof Paruch: Looking at this from bottom up, is...

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00:10:54.800 --> 00:11:00.310

Krzysztof Paruch: Trying to understand how you can, like, build an intuition and a framework

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00:11:00.320 --> 00:11:16.490

Krzysztof Paruch: to describe, those markets or those trading rules, if you really follow, kind of, like, the economic agenda or the publication stream. So, like, yeah, they're looking at that, Hansen 2003 publication, introducing,

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00:11:17.340 --> 00:11:20.329

Krzysztof Paruch: Scoring rules, information markets.

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00:11:20.950 --> 00:11:31.670

Krzysztof Paruch: and market scoring rules, and really understanding, kind of, kind of, like, how... how the things that he claims in this paper make sense, and how can they, can they be derived. So,

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00:11:31.720 --> 00:11:46.070

Krzysztof Paruch: What this led me to is, an investigation from the other side, so, like, a top-down approach, where looking more into, analysis of, convex analysis, and these kind of things, convex optimization.

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00:11:46.480 --> 00:11:53.040

Krzysztof Paruch: were... there is the possibility to, to derive, AMMs.

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00:11:53.630 --> 00:12:11.800

Krzysztof Paruch: top-down. So not from bottom-up, coming from, again, scoring rules, but more, like, from cost functions. And the whole, investigation here, was to kind of, like, yeah, bridge this gap, or to make an understanding how those two frameworks make sense.

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00:12:12.290 --> 00:12:25.820

Krzysztof Paruch: So this is what we will talk about, like, I will... I will give you a brief introduction, into where we stand. To everyone who has not been participating, I can even skim over the state of the publication, or, like, the state of the research that I'm... that I'm in.

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00:12:26.030 --> 00:12:42.030

Krzysztof Paruch: And then we will fast forward to one of the papers that I've selected as one helping me to kind of, like, bridge this gap. I don't know if I kind of, like, if it clicked fully, but we will see, hopefully, after the session, how much... how much of it really makes sense.

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00:12:42.250 --> 00:12:53.180

Krzysztof Paruch: And I've built some simulations as well, just to visualize this relationship. So this is what I have today, and let me share my screen and just,

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00:12:53.980 --> 00:12:56.750

Krzysztof Paruch: Yeah, I just, go along.

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00:12:59.060 --> 00:13:05.850

Krzysztof Paruch: Good. Okay, so you should now, and I hope this is right, you should be able now to see my HackMD document, is this correct?

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00:13:07.140 --> 00:13:07.810

Jamsheed Shorish: Okay.

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00:13:08.330 --> 00:13:17.400

Krzysztof Paruch: Perfect, okay, so I'm gonna be switching between those three documents, then I do also have a notebook here that I will open up.

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00:13:17.670 --> 00:13:21.130

Krzysztof Paruch: And, I might even come in with the paper.

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00:13:21.130 --> 00:13:40.230

Krzysztof Paruch: From the side. So there is a lot of stuff going on, and we will see how this all makes sense. So let's start with this paper, Hansen 2003, where he basically said, okay, we're looking at three different things. We're looking at scoring rules, we're looking at standard information markets, and we're looking at market scoring rules.

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00:13:40.230 --> 00:14:02.290

Krzysztof Paruch: And scoring rules are great to reward participants for truth-telling. So, like, this is a setup where you just basically say, you are asking people about their opinion, they are reporting something after the outcome, you can score their submission against the true outcome, and then you can pay them out based on the difference, or kind of, like, how close they were with their submission.

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00:14:02.940 --> 00:14:08.320

Krzysztof Paruch: So, so those are scoring rules. As particular.

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00:14:08.570 --> 00:14:24.799

Krzysztof Paruch: kind of scoring rules are proper scoring rules, where agents would even maximize their outcome only if they report their true beliefs. So this is even a nicer property, because in this scenario, we really can elicit truthful information. So we can... we have this truth-telling property.

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00:14:25.310 --> 00:14:37.139

Krzysztof Paruch: The problem of scoring rules, however, is, again, they fail at information pooling, so it's very difficult to have an aggregate market opinion, so it's basically just submitting different

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00:14:37.190 --> 00:14:50.489

Krzysztof Paruch: different answers, and then checking them against, kind of, what happened, and then scoring them, but not really being able to say, this is what the market believed, or, like, this was the aggregate opinion, this was the pooled opinion. So this is not happening very well.

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00:14:51.270 --> 00:15:09.210

Krzysztof Paruch: At the same time, you can look at information markets, where you would be trading beliefs, and they do better at aggregating information, but they have limitations such as the thin market problem and irrational participation problem, which means that they don't perform well if there are not many people trading.

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00:15:09.210 --> 00:15:22.640

Krzysztof Paruch: And at the same time, people don't... are not really incentivized to... to really participate, except they have hedged against their risks. So, like, there is... there is no real motivation to kind of, like, engage in those markets.

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00:15:22.700 --> 00:15:34.670

Krzysztof Paruch: And what you see on this diagram here is that if you, if you plot the estimates per trader, and you look at the accuracy, then you would see that information markets are

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00:15:34.670 --> 00:15:47.200

Krzysztof Paruch: are good here, but very poor here. So if you have a lot of estimates per trader, so there is a thin market, basically, there are not many people in, and one trader submits many, many opinions.

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00:15:47.200 --> 00:16:06.160

Krzysztof Paruch: This is where you have this thin market problem. And at the same time, scoring rules fail at opinion pooling. So, in the other scenario, where you would be looking at very little information coming from one single trader, it's very difficult to aggregate them and get this accuracy.

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00:16:06.410 --> 00:16:14.680

Krzysztof Paruch: And what Hansen proposes, then, is the out-of-market scoring rules, which form the missing link between scoring rules and information market.

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00:16:14.680 --> 00:16:30.649

Krzysztof Paruch: And what happens there is you take a proper scoring rule and turn it into an always-liquid market maker. So this is what you, if you've participated last time, you are familiar with this document, which is basically how this works.

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00:16:30.650 --> 00:16:32.679

Krzysztof Paruch: So, so it describes...

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00:16:32.710 --> 00:16:57.640

Krzysztof Paruch: both what cost functions are here in this context, and also it describes what information aggregation mechanisms are, how scoring rules work. So, like, you start with the scoring rule as P , you define what proper scoring rules are, you get into strict scoring rule, you have two examples, the quadratic scoring rule and the logarithmic scoring rule, which is something that we will encounter quite a bit in

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00:16:57.640 --> 00:16:59.060

Krzysztof Paruch: In this presentation.

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00:16:59.360 --> 00:17:22.669

Krzysztof Paruch: And then you, you kind of, like, here we discuss those, those problems and those, those difficulties with them. In the next section, I discuss the standard information markets, where we can trade the leaves. We typically encode this by a Bernoulli security, where we have a security that pays 1 if some outcome is realized and pays 0 otherwise.

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00:17:22.670 --> 00:17:42.929

Krzysztof Paruch: So this is typically how shares are, encoded in prediction markets. You basically buy a share of one outcome, and you say, okay, this is what's happening here. If this event materializes, I will win, because my payout is going to be won. In all the other cases, my payout is going to be zero. And by trading those, those,

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00:17:43.040 --> 00:17:53.240

Krzysztof Paruch: securities, you get to, kind of, like, price discovery and, information aggregation, but this doesn't work super well, in,

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00:17:53.460 --> 00:17:57.899

Krzysztof Paruch: In the case where you have this, in-market, right?

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00:17:58.280 --> 00:18:09.139

Krzysztof Paruch: So, so then, market scoring rules. And market scoring rules, are defined as follows. You start with a proper scoring rule, and then you...

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00:18:09.400 --> 00:18:25.140

Krzysztof Paruch: You take this and you allow people to, have a collectively, held, by the market, a collectively held, probability distribution. So this is what's actually happening. So you have a market forecast.

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00:18:25.140 --> 00:18:49.940

Krzysztof Paruch: And anyone is allowed to update this forecast by trading against the market, so, like, shifting their beliefs, by paying something, so the cost for the shift is defined as this market scoring rule. So, like, this is exactly how much it would cost a trader to move from one inventory to another. And the market has to remember those,

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00:18:49.940 --> 00:18:56.850

Krzysztof Paruch: Those shifts, those transitions, such that when the outcome is realized, it can be paid out.

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00:18:56.910 --> 00:19:01.659

Krzysztof Paruch: And all the, all the claims can be, can be resolved.

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00:19:02.290 --> 00:19:04.150

Krzysztof Paruch: And,

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00:19:04.370 --> 00:19:16.079

Krzysztof Paruch: what Hansen further claims in his paper is, is how this works. So, taking the logarithmic market scoring rule as one special case, which is one that is very prominently used.

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00:19:16.080 --> 00:19:26.089

Krzysztof Paruch: He shows and he defines, what this actually means, and how, how this market scoring rule, is... is the cost function of

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00:19:26.450 --> 00:19:40.210

Krzysztof Paruch: of Q . And now what you see, happening is this shift from, kind of like a state update, which is basically changing... changing beliefs of a probability of particular outcomes.

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00:19:40.710 --> 00:19:54.799

Krzysztof Paruch: to something that indexes Q , which is a change of inventory. So, for me, the key question here was, how do you map between those two spaces? How do you map between updating probabilities

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00:19:54.800 --> 00:20:10.829

Krzysztof Paruch: towards, towards how do you map between updating, inventory, because this is what is de facto happening, right? So, like, a change, a trade in the market, de facto is a change in holdings of shares of market participants.

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00:20:11.120 --> 00:20:19.790

Krzysztof Paruch: And, what the result of this is, and what the consequence of all of this interaction is, that,

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00:20:20.290 --> 00:20:37.749

Krzysztof Paruch: You can do this, and you have to pay particular prices for doing this, and if you finalize this step, then the holdings, the prices of the holdings will be exactly the probabilities of those events.

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00:20:37.750 --> 00:21:01.109

Krzysztof Paruch: as aggregated by the market, which is kind of like this beautiful thing and this very strong... this very strong argument, but in this argumentation, kind of like there's a lot hidden under the hood. So for me, the big question was, okay, if you start here, if you look at this market scoring rule, and if it's the logarithmic market scoring rule, which is defined as $\log Q$ minus $\log P$,

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00:21:01.110 --> 00:21:07.390

Krzysztof Paruch: how... how the hell do you... do you say, then, okay, that the cost function is then $B \log$ the sum of this, right?

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00:21:07.550 --> 00:21:16.579

Krzysztof Paruch: And in order to... in order to resolve this, I was looking into... into this publication here, which is, which is the one that I'm gonna show

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00:21:16.890 --> 00:21:41.689

Krzysztof Paruch: in a minute, but from here, like, if you have the cost function, it's pretty straightforward to derive the prices. So, like, the partial derivatives of the cost function are the prices. So you can... you can say that the prices of each share vector to have an infinitesimal change in those holdings, would be the partial derivatives in each of the directions of the... of the

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00:21:41.940 --> 00:22:01.700

Krzysztof Paruch: inventory, which... which gets you to this, right? So, so yeah, so this is, this is basically where, where I stopped when I kind of read Hansen, bottom-up, right? Scoring rules, information markets, market scoring rules. And the question was, okay, how can you build to actually, like.

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00:22:01.810 --> 00:22:21.299

Krzysztof Paruch: something that encompasses all of this, which is a nice metaphor, because we will actually work with the complex hull now, so we will actually try to build this argument that will help us understand how those prices and how those hyperplanes, actually, that support this cost function

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00:22:21.300 --> 00:22:26.770

Krzysztof Paruch: Are derived. Let me stop at this moment, maybe there is some feedback or some questions, some comments.

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00:22:31.830 --> 00:22:32.800

Jamsheed Shorish: So...

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00:22:32.800 --> 00:22:34.090

Krzysztof Paruch: One...

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00:22:34.160 --> 00:22:46.560

Jamsheed Shorish: It's not really a question, more kind of an observation that this literature is, in a way, it's a superset above some of the traditional finance work on pricing asset-backed securities.

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00:22:46.890 --> 00:23:02.810

Jamsheed Shorish: So there is a fair amount of overlap in terms of how the prices themselves reflect probabilities in the literature which uses what are called state prices. I don't know if you've come across that already yet in the work that you've looked at before.

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00:23:02.960 --> 00:23:18.920

Jamsheed Shorish: That kind of... it's like a deflator of the prices, and it allows you to do things more general than just having aerode Securities giving just a 1 and 1 in one particular location and a 0 otherwise, which is why you can get the normalization of the price to be between 0 and 1 as a probability.

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00:23:18.950 --> 00:23:30.129

Jamsheed Shorish: But it actually allows you to look at the way in which the realizations that will occur from these things that are continually uncertain still gives you a consistent pricing kernel.

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00:23:31.020 --> 00:23:44.310

Jamsheed Shorish: So, I'm trying to think now, and as we go through the talk, I'll try to think of a good reference that might be good to kind of delve into, because once you start associating probabilities with prices, you are in the state price universe for finance.

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00:23:44.630 --> 00:23:58.770

Jamsheed Shorish: And I think it's... would be important to make that connection between them, because in traditional markets, traditional financial markets that clear with state prices are going to have a representation in the... in the market making, the automated market maker that you create.

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00:24:03.300 --> 00:24:13.300

Krzysztof Paruch: Does this, correspond to, kind of, like, the ambition of Hansen, and also in the other literature, to kind of, like.

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00:24:13.710 --> 00:24:21.659

Krzysztof Paruch: approach, combinatorial markets, so... so markets... no, this is not...

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00:24:21.660 --> 00:24:27.690

Jamsheed Shorish: No, no, although it can be. It can be used in that context. This is much more, in a way.

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00:24:27.880 --> 00:24:41.829

Jamsheed Shorish: I don't want to call it restrictive, but it's much more limited to the way in which you pass from an objective probability measure to a probability measure that, under its association, you can price everything as if you are a risk-neutral individual.

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00:24:41.910 --> 00:24:50.060

Jamsheed Shorish: Here, the solutions that I've... as I was reading through from Hansen and Abernathy, you know, the main strong results are for risk-neutral investors.

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00:24:50.740 --> 00:24:57.940

Jamsheed Shorish: Across the different objective space, so you can actually match these probabilities up as just weighted averages, essentially, across those values.

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00:24:57.940 --> 00:25:12.110

Jamsheed Shorish: And that's super useful, right? Because that allows you then to associate with a particular payoff the multiple of the probability of the outcome occurring. But this is a much more expanded version of that, where it says, okay, look, let's actually understand

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00:25:12.110 --> 00:25:16.170

Jamsheed Shorish: For any kind of probability distribution, for any type of risk preferences.

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00:25:16.230 --> 00:25:31.370

Jamsheed Shorish: that may be held by the individuals trading in the marketplace, that they can all agree that there exists a risk-neutral probability measure, and that under that measure, people would behave as if they were trading according to that reflection. And that's what state prices do.

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00:25:31.910 --> 00:25:42.490

Jamsheed Shorish: So there's a very close connection between this assessment of what is being done on this automated, this AMM side, and what traditional finance has done in the development of state pricing.

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00:25:42.490 --> 00:25:56.870

Jamsheed Shorish: So it's worth kind of making some glue to stitch those together, I think, because that's something, at least, if nothing else, you know, a referee will say, hey, what about state prices? But also, it may actually shed some light into how to make this general enough that you can use any kind of risk preferences

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00:25:56.870 --> 00:26:03.330

Jamsheed Shorish: to price the commodities. And that could become important in situations where you have local failure of,

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00:26:03.430 --> 00:26:08.130

Jamsheed Shorish: Of convexity of the cost function, or cost functions which are not actually continuous.

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00:26:08.740 --> 00:26:27.369

Jamsheed Shorish: Because that's, of course, one of the ways in which you need to kind of build this association with the cost function is that you need to be able to take, like, a spot price, an instantaneous price as a derivative. There may exist markets for which that cost function isn't continuous, and you have to make it continuous yourself, but that may open up all kinds of other problems.

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00:26:27.390 --> 00:26:33.569

Jamsheed Shorish: I would look into that literature just to at least make sure that it's part of the reference list, but it may also shed light on more details.

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00:26:34.390 --> 00:26:43.389

Krzysztof Paruch: Okay, so the argument here is that, IMMs with complex cost functions are just a subset of,

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00:26:44.470 --> 00:26:52.799

Krzysztof Paruch: Trading possibilities for more general Right. Strategic players who are not risk neutral.

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00:26:53.360 --> 00:27:06.900

Jamsheed Shorish: That's correct, yeah. And for much more general probability spaces for which the events are occurring, you can, for example, create subsets of the event space, and then assign probability measures on those.

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00:27:06.900 --> 00:27:24.909

Jamsheed Shorish: As opposed to having, strictly speaking, perfect aerod security, which, you know, it pays one in the unambiguous, everyone agrees that this event occurred, and zero otherwise. You can use a state price representation to have, well, we don't know exactly what happened, but we're pretty sure it's... we know that it was within this particular set of things that maybe has occurred.

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00:27:24.980 --> 00:27:35.910

Jamsheed Shorish: And that allows you to have learning, and allows you to have heterogeneous updates of the expectations that you might have, given risk preferences that themselves are very different across the population.

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00:27:36.160 --> 00:27:39.260

Jamsheed Shorish: I'll try to find... Yeah.

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00:27:39.630 --> 00:27:47.680

Jamsheed Shorish: Yeah, give me a moment, I'll try to pull the book off and pass it on to you, or at least show it in the camera and put it in the chat, but I can't remember exactly.

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00:27:48.410 --> 00:27:50.450

Krzysztof Paruch: Perfect. Sounds good. Okay.

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00:27:50.990 --> 00:28:06.800

Krzysztof Paruch: Let me move on. So what we want to... yeah, so maybe here. So this is what, what we've kind of, like, used as the motivation. And now, when we... when we look at the main questions is, okay, if we have such a scoring rule, how can we derive a cost function?

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00:28:07.180 --> 00:28:11.110

Krzysztof Paruch: Such a trading against fees actually equivalents to the market scoring rule updates.

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00:28:11.270 --> 00:28:25.549

Krzysztof Paruch: And market scoring rule state is a probability vector P , cost function state is an outstanding shares Q . What is the mapping between these state spaces? So, so this is basically the definition of, okay, where I left off.

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00:28:26.540 --> 00:28:43.260

Krzysztof Paruch: Yeah, and then, we can look at, at this definition, which is, what I've shown in the paper. So, like, we can skip this section, because this is what I've already showed you. So, so,

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00:28:44.530 --> 00:28:48.870

Krzysztof Paruch: Yeah, let's just skip it, because this is directly from this paper.

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00:28:49.300 --> 00:29:07.370

Krzysztof Paruch: And then we can look into this Aberlatti paper that basically provides a general framework to design markets from top down, and it allows me to... to provide... to prove that any market that is satisfying a set of intuitive conditions

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00:29:07.870 --> 00:29:20.180

Krzysztof Paruch: Must prize security via a convex cost function, excluding what you just said, jump sheet, kind of like the special cases, like, assuming that this statement is only true for risk-neutral traders.

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00:29:20.510 --> 00:29:37.460

Krzysztof Paruch: And that you can construct this function that is representing the market via conjugate duality. So, let us at first maybe look into those conditions. So, like, what are the desirable conditions that we want to have, for which we would like to derive this framework?

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00:29:37.800 --> 00:29:47.240

Krzysztof Paruch: Condition number one is path independence. So, if we have any sequence of trades entering the market, and we have any trader coming in.

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00:29:47.950 --> 00:29:58.159

Krzysztof Paruch: They should be able to retrieve a cost of any bundle, given a previous trade sequence, and it should be irrespective of the ordering of this trade sequence.

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00:29:59.000 --> 00:30:11.460

Krzysztof Paruch: And, if we... if we take this condition, we already see that, prices can be represented as a cost function, which is theorem number one in this paper.

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00:30:13.040 --> 00:30:20.490

Krzysztof Paruch: In... but... We don't only want to have, the representation by a cost function.

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00:30:20.710 --> 00:30:34.589

Krzysztof Paruch: we want a little bit more. So we want the market to also aggregate traders' beliefs into one accurate prediction, or we want the market to also pool information. As we mentioned before, this is what we are interested in, so we want this property.

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00:30:34.920 --> 00:30:50.450

Krzysztof Paruch: And in order to achieve that, we need to define five more conditions. So, we have condition number 2, which is a well-defined gradient of C . Again, C has to be continuous and differentiable everywhere.

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00:30:50.800 --> 00:30:55.660

Krzysztof Paruch: And I assume that, again, Japit, that all of those conditions are true for this

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00:30:55.880 --> 00:30:58.430

Krzysztof Paruch: Special case of risk neutrality.

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00:30:59.130 --> 00:30:59.970

Krzysztof Paruch: It would...

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00:30:59.970 --> 00:31:11.280

Jamsheed Shorish: Typically be so, yeah. I mean, the, I think probably the trick here is that if you want sharper results, you actually need more. You would need it to be at least C^2 .

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00:31:11.750 --> 00:31:29.870

Jamsheed Shorish: I'm pretty sure the second derivative is required for that, especially when you look at the rate of curvature of the convexity. But this is where you might have, for example, the possibility that you can't find the continuous convex cost function.

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00:31:30.040 --> 00:31:33.390

Jamsheed Shorish: Because the idea of the cost function is it's essentially like a,

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00:31:33.900 --> 00:31:51.009

Jamsheed Shorish: It's like a group in the sense that what it's doing is it's taking the amounts that you could be buying and saying, well, the costs themselves should be the sums of the costs associated with buying those pieces individually. So, like, the order doesn't matter in a way. And,

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00:31:51.190 --> 00:31:58.589

Jamsheed Shorish: Whether or not that's actually true in markets that are indivisible, that's something where you have holes, right? You have certain amounts that you cannot buy.

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00:31:58.790 --> 00:32:07.819

Jamsheed Shorish: Or they become lumpy, like in real estate, where you can't buy half a house, for example. That would end up causing you to have trouble with a gradient approach.

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00:32:10.100 --> 00:32:10.920

Krzysztof Paruch: Correct.

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00:32:12.020 --> 00:32:26.879

Krzysztof Paruch: Condition number three, information incorporation. So, there should be only one way to express one's information, which is, critical. So, like, there shouldn't be a possibility for someone to kind of, like, get the same result if, they...

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00:32:27.100 --> 00:32:32.589

Krzysztof Paruch: By opposite beliefs, so to say. Which means that,

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00:32:33.010 --> 00:32:45.420

Krzysztof Paruch: the purchase of a security bundle should never cause the market to lower the price of the security bundle. So, kind of like this, there are two ways of saying this, so either you say this, where you basically always increase if you buy it, you increase the price.

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00:32:45.440 --> 00:32:56.430

Krzysztof Paruch: Or you say there is no way to, kind of, like, hedge again, so, like, arbitrage away, kind of, like, your beliefs by saying, okay, I'm buying two opposite positions, which would, allow me to make, a win guarantee.

196

00:32:56.430 --> 00:33:08.119

Krzysztof Paruch: Which is also the condition number 4, which is the non-arbitrage position, so you cannot buy any bundle that gives you profit regardless of the outcome.

197

00:33:08.200 --> 00:33:23.200

Krzysztof Paruch: So, so this is important. And then expressiveness just means that you can buy any tiny amount of bundles as you wish. So, which would result in... in setting the market prices to actually reflect their beliefs

198

00:33:23.200 --> 00:33:30.890

Krzysztof Paruch: About the expected payoffs of each security, so they can move in, you know, the tiniest epsilons across the space.

199

00:33:32.160 --> 00:33:45.589

Krzysztof Paruch: And as a consequence of those conditions, the pricing mechanism is always described precisely in terms of a convex cost function C , and the set of reachable prices must be a convex hull of the pay of vectors for each outcome.

200

00:33:45.590 --> 00:33:54.120

Krzysztof Paruch: So this is what, what we will be, this is, what we will be looking at in the simulations. So if you look at the...

201

00:33:54.790 --> 00:33:59.320

Krzysztof Paruch: At the cost functions and supporting hyperplanes.

202

00:33:59.320 --> 00:34:24.110

Krzysztof Paruch: And you define any convex function, and you look at how you can build a space between... that represents both the quantities and the prices, you would end up in the representation that prices are the gradient of this cost surface, which allows you to define a plane formula of the hyperplane that you,

203

00:34:24.110 --> 00:34:33.419

Krzysztof Paruch: Built out of this touchpoint. And this gets you to... let me show you the nice thing, because it's what you're actually looking at.

204

00:34:33.540 --> 00:34:49.150

Krzysztof Paruch: this is the cost surface with the supporting

hyperplane in the special case of the logarithmic market scoring rule. So you would have, here the... your... your inventory of Q1, your inventory of Q2, and in this,

205

00:34:49.570 --> 00:35:06.599

Krzysztof Paruch: case, the inventory Q3 is set to zero, because it's not representable otherwise, and then we have this point on the cost function, and then you see that the resulting hyperplane is exactly here, right? So, like, you have this, this tangent plane

206

00:35:06.630 --> 00:35:21.619

Krzysztof Paruch: That is... is touching this point, and it's beneath the... the cost function everywhere, except for... for this one point where it touches it. And, this is even nicer to show on this,

207

00:35:21.620 --> 00:35:27.100

Krzysztof Paruch: In a, you know, like, animated version, where you would see how moving on this plane

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00:35:27.100 --> 00:35:51.779

Krzysztof Paruch: on this cost function actually behaves. So, so you, you move with this red, red point, obviously, across the cost function, and this changes the inventory, Q1, Q2, and, accordingly, this changes also the, the gradient, or the, or the steepness, of those functions. And in each of these points, I think I have it somewhere here, let me see...

209

00:35:51.990 --> 00:36:10.040

Krzysztof Paruch: you can build errors, so you can actually represent the partial derivatives, of, of this point in both directions, Q1, Q2, which would exactly be the prices to move in this direction. I think I have it as the next...

210

00:36:10.270 --> 00:36:12.660

Krzysztof Paruch: Or one of the next,

211

00:36:14.020 --> 00:36:21.890

Krzysztof Paruch: No, not really, I think. But, I might have it in this representation here.

212

00:36:22.590 --> 00:36:42.470

Krzysztof Paruch: Which is... yeah, so here I have it. I don't know

where I have this from, but this is, the cost surface with the supporting hyperplane and the price vectors, which would be exactly those, derivatives, the partial derivatives in those respective directions, and you can get, you can get those, those prices here.

213

00:36:43.360 --> 00:36:44.760

Krzysztof Paruch: And...

214

00:36:44.940 --> 00:36:58.469

Krzysztof Paruch: There are some conditions, obviously, so you see that, like, if you have price 1, price 2, you have an implied price 3, like, it has to add up to 1 in this case, although it is, in my example, not numerically derived, but there is no other way.

215

00:36:58.920 --> 00:37:00.969

Krzysztof Paruch: For it to, to fit in.

216

00:37:02.570 --> 00:37:05.979

Krzysztof Paruch: Okay. So, so this is the...

217

00:37:06.910 --> 00:37:14.659

Krzysztof Paruch: The cost function, you could... this is the way how you would show that you can actually build

218

00:37:14.890 --> 00:37:20.840

Krzysztof Paruch: the, this... No, no, sorry. You can build this...

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00:37:22.110 --> 00:37:32.049

Krzysztof Paruch: host surface by supporting hyperplanes. And what I have simulated here is,

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00:37:33.040 --> 00:37:46.169

Krzysztof Paruch: If you just... if you have the true cost surface, and you are looking at an approximation from each point, you would be able to make a comparison between what the

221

00:37:46.170 --> 00:37:56.190

Krzysztof Paruch: gap is that you have from, from calculating the hyperplane and the kind of, like, the support that you have towards the true

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00:37:56.200 --> 00:37:59.430

Krzysztof Paruch: The true cost function. Is it this...

223

00:37:59.480 --> 00:38:01.990

Krzysztof Paruch: This... this is what it should be.

224

00:38:02.720 --> 00:38:22.390

Krzysztof Paruch: Right. So, so here the gap, of CQ minus, C star Q , that, that shows, okay, you... you can actually derive those, this, this approximation of the cost function through, through a , through, through the convex hull of.

225

00:38:22.610 --> 00:38:26.710

Krzysztof Paruch: Of the, of the function.

226

00:38:28.150 --> 00:38:41.190

Jamsheed Shorish: Just to clarify, these are the tangent planes generated from the payoff vectors, right? So that you're actually looking at given information that's given by the structure of the securities that are present.

227

00:38:41.190 --> 00:38:50.220

Jamsheed Shorish: So different payoffs would have created different tangent planes, which then would be aggregated to a different convex hull for the cost function to be in.

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00:38:51.160 --> 00:38:51.820

Krzysztof Paruch: Right.

229

00:38:56.130 --> 00:39:11.160

Krzysztof Paruch: Okay, but this would be just, yeah, kind of, like, showing that this really works, so, like, that you... that you can represent this by this calculation. The convex function is the maximum of all of those, realizations.

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00:39:13.380 --> 00:39:27.779

Krzysztof Paruch: Nice. And then, what you can do, and this is, basically what now, Abernetti, Chang, and, Walken say is, if you, if you have this, this

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00:39:27.900 --> 00:39:41.110

Krzysztof Paruch: those conditions, you can design the cost function via conjugate duality. So... so this is how, how this, the cost function, has to look like this.

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00:39:41.390 --> 00:39:56.569

Krzysztof Paruch: And, in the pa... in the paper, they derive, how you can represent... how you can represent this. So... so you would start with the augmentation. Any... any convex closed differential function C can be written in the form

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00:39:56.570 --> 00:40:03.749

Krzysztof Paruch: CQ is just a premium overall axis in the domain of R , X times Q minus R of X ,

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00:40:03.790 --> 00:40:13.309

Krzysztof Paruch: were, where you just choose a convex function R , which is called the conjugate of C .

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00:40:13.420 --> 00:40:25.380

Krzysztof Paruch: And, then you can, you can make the gradient of C to be expressed in terms of this conjugate, so you can say, the gradient of C is the

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00:40:25.620 --> 00:40:27.930

Krzysztof Paruch: the X that is maximizing this?

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00:40:28.400 --> 00:40:44.630

Krzysztof Paruch: And, what it spits out is, is this sequel X , which is, the supremum of all, X s on the domain that you have restricted R to, so kind of like all the points that you would be looking at.

238

00:40:45.150 --> 00:40:49.550

Krzysztof Paruch: And this gives you the complex cost function market maker.

239

00:40:50.520 --> 00:40:56.220

Jamsheed Shorish: Let's, just as a quick pause here, just to kind of, like, again, sort of fix some intuition.

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00:40:56.760 --> 00:41:04.120

Jamsheed Shorish: First, from the perspective of somebody creating this, where they've walked in and seen all these securities lying around with different payoff vectors.

241

00:41:04.310 --> 00:41:12.849

Jamsheed Shorish: does it make sense that you can choose, when you say here, an appropriate conjugate function R , can you choose R independently?

242

00:41:13.090 --> 00:41:24.479

Jamsheed Shorish: of the payoff vectors of the securities? I see the restriction of the domain, but can you actually choose the function independently of some structure in the underlying securities?

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00:41:25.940 --> 00:41:38.719

Krzysztof Paruch: What they say in the paper is that there are some, candidate R s, so, like, there is... there is a particular way to choose, to choose this R , to... to achieve, to, to achieve something. So, so this is the argumentation.

244

00:41:38.720 --> 00:41:53.500

Jamsheed Shorish: I feel like there must be an algorithm, right? Because I can think of a very degenerate economy where, like, all of the securities pretty much pay almost the same amount for almost the same events.

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00:41:53.870 --> 00:41:58.609

Jamsheed Shorish: So they're, like, in some sense, in an abstract sense, collinear with each other.

246

00:41:59.030 --> 00:42:04.459

Jamsheed Shorish: And there, you really have to make sure you're choosing the correct R , because you can't choose

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00:42:04.640 --> 00:42:11.559

Jamsheed Shorish: An R which expands and explores dimensions of payoffs, because the convex hull is super narrow.

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00:42:12.290 --> 00:42:16.030

Jamsheed Shorish: So there has to be some way of connecting the choices of our

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00:42:16.220 --> 00:42:32.790

Jamsheed Shorish: to the underlying span of the payoff vectors, and I didn't see that in the Abernathis paper. I couldn't really tell where that was coming from. It was sort of posited as there exists kind of an R , so I didn't know if you'd seen anything that had really been a little bit more in the weeds on how to construct

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00:42:32.790 --> 00:42:40.340

Jamsheed Shorish: That particular thing to make this hyperplane argument, really implementable, like, as a methodology.

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00:42:41.440 --> 00:42:50.420

Krzysztof Paruch: So this is one thing. The other thing that I mentioned is that there might be some problems at the boundaries, right? So that... so that you... so that you... you get to...

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00:42:50.540 --> 00:42:57.240

Krzysztof Paruch: A situation where all... all changes would be... would be at the limit, priced.

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00:42:57.890 --> 00:43:02.809

Krzysztof Paruch: the same, which is also something that you, that you wouldn't want, right? So.

254

00:43:04.210 --> 00:43:22.979

Jamsheed Shorish: Because you've got, like, this... the program you've got is really cool, right? Because you're trying to build from the top down and the bottom up to try to have them meet in the middle, where you can get these scoring rules to be supported by cost functions in a way that is consistent. And one way to do that is to say the cost functions are the logarithmic scoring rule, and then you build the cost function that way and show X , you know, X post that that exists.

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00:43:22.980 --> 00:43:25.270

Jamsheed Shorish: But the thing is, if I handed you

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00:43:25.360 --> 00:43:31.399

Jamsheed Shorish: the logarithmic scoring rule, and a really weird set of securities, I'm not sure that construction will fit.

257

00:43:31.740 --> 00:43:49.110

Jamsheed Shorish: You know what I mean? You may end up with, like, the cost function just can't be supported from below, because the payoff system just isn't... isn't rich enough, or expressive enough, and I didn't see the conditions under which the expressivity is necessarily present, with those five conditions that are being used by Abernathy as their, like.

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00:43:49.300 --> 00:43:52.470

Jamsheed Shorish: presupposing what you need to have. So, just something I was wondering.

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00:43:52.470 --> 00:44:01.659

Krzysztof Paruch: So what... no, so what you're saying is that they impose that for those five... you can find a cost function for any R ,

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00:44:01.800 --> 00:44:05.560

Krzysztof Paruch: If it satisfies those five conditions, but they don't tell you how.

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00:44:06.130 --> 00:44:19.080

Jamsheed Shorish: Yeah, I don't think so. That's my... that's my question. I'm wondering if that's, like, an external thing that still needs to be, like, an art form, where even when you have those five conditions, because those five conditions really are more on, like, the kinds of admissible cost functions.

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00:44:19.760 --> 00:44:24.510

Jamsheed Shorish: So if you've given yourself one, like the logarithmic rule, which you know already satisfies these.

263

00:44:24.730 --> 00:44:41.030

Jamsheed Shorish: you've still left the underlying state space, the underlying space of the relationships between what it is that you would have as a particular payoff and what you would expect to see, given different types of events, kind of up to... up to nature, right? And so the question is kind of like.

264

00:44:41.030 --> 00:44:54.839

Jamsheed Shorish: Does nature behave well enough to make that very nice cost function that's found stitched together in such a way that you can use that expressivity of the expansion of a cost function, convex cost function, into its conjugate form?

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00:44:56.030 --> 00:45:07.810

Jamsheed Shorish: That's the thing I just... I don't know, because it all boils down to selecting that R functional, that strictly convex function, R. R has very strong properties, because it has to be strictly convex, and I don't know if that's true.

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00:45:08.640 --> 00:45:26.190

Krzysztof Paruch: Right, I mean, I... yeah, conceptually, I didn't even get that far in the... in this paper, so... so, yeah, I wasn't wondering about those... those questions. I was wondering about more fundamental questions, but, yeah, I... so, just to answer your question, I haven't found an answer to this.

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00:45:26.190 --> 00:45:32.009

Jamsheed Shorish: No worries, yeah. I'm curious, and it's interesting. It's an interesting question just because of the fact that it's

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00:45:32.670 --> 00:45:47.850

Jamsheed Shorish: Securities markets are highly incomplete, which, of course is one of the things that's kind of, like, being discussed, is like, how do we want to improve ourselves to be able to, like, get away with being able to do more things with less? Like, having a fewer set of sets of instruments?

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00:45:47.850 --> 00:45:56.659

Jamsheed Shorish: But they're also odd. They also duplicate a lot. There's a lot of places where the manifolds collapse, or they have cusps, or weird corners.

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00:45:56.660 --> 00:46:05.640

Jamsheed Shorish: And it's hard to kind of get the feel for the justification that you're building a lot of structure in the cost function from above, which makes perfect sense.

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00:46:05.640 --> 00:46:20.200

Jamsheed Shorish: But that you might not necessarily have the market structure that will... in the state space, that will actually support that in the end. So, it's just something to stick a pin into and see whether or not there is, like, a means of ensuring that you're guaranteed to be able to make that program work.

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00:46:21.290 --> 00:46:22.090

Krzysztof Paruch: Right.

273

00:46:24.120 --> 00:46:39.539

Krzysztof Paruch: Okay, and then finally, so, what I really wanted to understand is, how does this look for the logarithmic market scoring rule, and how to derive this? So, what are the steps you need to take just to kind of get this calculation? And,

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00:46:39.540 --> 00:46:56.390

Krzysztof Paruch: I did this, so, like, I went through, kind of, like, the derivation, which is straightforward. So, like, if you start up this cost function, CQ , you can define the conjugate, which is exactly this. So, C^* , in this case, is the supremum.

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00:46:56.780 --> 00:47:00.589

Krzysztof Paruch: Overall, Q is P times Q minus this cost function.

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00:47:01.250 --> 00:47:15.159

Krzysztof Paruch: And then there is just one trick you need to do, which is rescaling. So if you do this, you replace, UI , QI over B with UI , and you get to this representation.

277

00:47:15.480 --> 00:47:26.449

Krzysztof Paruch: You can even look only at the supremum here inside, so even, like, taking out B , just to make it simpler so that you say that the conjugate is B times this V .

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00:47:27.000 --> 00:47:36.709

Krzysztof Paruch: And then what you do is you look at conditions that you need to meet, such that the supremum is finite, and what immediately pops out are two conditions that,

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00:47:37.560 --> 00:47:55.429

Krzysztof Paruch: make you land on the probability simplex, which is basically the sum of all P s has to be 1, and all P s have to be bigger than 1. And if you do this, then you get to kind of, like, all admissible P 's to have this condition.

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00:47:55.600 --> 00:48:01.890

Krzysztof Paruch: And, you can solve for the supremum for all of the P 's that are here, which is

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00:48:02.040 --> 00:48:12.730

Krzysztof Paruch: Just the partial derivative of, of this function to all use, which are now kind of like proxies for your quantities, like the substituted quantities.

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00:48:12.900 --> 00:48:17.930

Krzysztof Paruch: And it gets you to this condition, which you can set to zero, so the prices are exactly this.

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00:48:18.160 --> 00:48:32.990

Krzysztof Paruch: And, the way to satisfy this is by setting U to be this. You can plug this back into the objective, and you get, to this representation, which, after kind of, like, multiplying with B , you get to the complex conjugate looking like this.

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00:48:33.230 --> 00:48:51.910

Krzysztof Paruch: Which is, B , if you run a probability simplex, it's exactly this, and otherwise, you can make it go to infinity by just choosing, choosing, either by adding, a constant, or by, letting,

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00:48:52.020 --> 00:49:04.239

Krzysztof Paruch: in this case, so you can add a constant here just to, like, make this explode, or in this case, you can make U go to minus infinity, and then you would be also

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00:49:04.410 --> 00:49:17.469

Krzysztof Paruch: you would also blow up the supremum. So in both cases, you end up with infinity as a result, but if you're on the probability simplex, you get, like, one particular complex conjugate that you can work with, and this is the one that you have.

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00:49:18.100 --> 00:49:21.110

Krzysztof Paruch: Yeah, so, so...

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00:49:21.370 --> 00:49:37.819

Krzysztof Paruch: kind of, like, for the logarithmic scoring rule, deriving this, and now, finally, I can show you how this would look if you are looking at a visualization of this. So, to answer the question, like, what is the connection between the primal and the dual space.

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00:49:37.820 --> 00:49:49.150

Krzysztof Paruch: for LMS, our market makers, we have this primal space, which is the quantity space, and we have the dual space, which is the belief space, that we have to just define as the probability simplex.

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00:49:49.190 --> 00:49:57.349

Krzysztof Paruch: And then we also have this C star to be defined like this. And in our three-dimensional case, we just add up from 1 to 3.

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00:49:57.580 --> 00:50:04.189

Krzysztof Paruch: And then you can, just show what's happening side by side. If you're trading on,

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00:50:04.400 --> 00:50:09.959

Krzysztof Paruch: On this, curve, you move the hyperplane, and you also kind of, like,

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00:50:10.650 --> 00:50:18.420

Krzysztof Paruch: move within, kind of, like, the bounds of the probability simplex. So this is what you would see as a visualization here.

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00:50:18.940 --> 00:50:20.790

Krzysztof Paruch: And... and...

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00:50:21.290 --> 00:50:31.719

Krzysztof Paruch: this is where I've gotten, so this is kind of, like, how far I've gotten in understanding all of this. I'm not quite sure if I... if I,

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00:50:31.820 --> 00:50:46.059

Krzysztof Paruch: managed to answer the question that I set out to answer, which is, how can you now make this, like, intuitive connection between, market scoring rules and cost functions, like, very explicit, like.

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00:50:46.210 --> 00:50:57.530

Krzysztof Paruch: But I feel like I've made a lot of progress in between, and it makes sense, but again, it hasn't clicked fully yet. So, yeah, maybe using the conversation today

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00:50:58.100 --> 00:51:05.030

Krzysztof Paruch: Just to reassure myself that this all makes sense, and there are no... blank spot.

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00:51:06.750 --> 00:51:24.699

Jamsheed Shorish: Well, one of the things I think... first of all, I think it's great. I think these simulations are super useful to get a visualization of what's happening at each point as the two systems themselves are representing each... a different aspect of the information aggregation and the truth-telling, the revelation part of the system.

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00:51:25.190 --> 00:51:26.800

Jamsheed Shorish: Probably the...

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00:51:27.220 --> 00:51:34.579

Jamsheed Shorish: The thin edge of the wedge for another way to expand beyond, like, the logarithmic scoring rule in particular.

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00:51:34.810 --> 00:51:36.440

Jamsheed Shorish: Is to take that

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00:51:37.040 --> 00:51:45.349

Jamsheed Shorish: really that complex dual seriously, and that definition of a negative entropy seriously, because that's information.

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00:51:45.930 --> 00:51:51.250

Jamsheed Shorish: And so, that essentially is where the information theory intersects with

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00:51:51.380 --> 00:51:58.030

Jamsheed Shorish: The processing of the elicitation of truth-telling, which is what the scoring rule itself is doing.

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00:51:58.090 --> 00:52:12.009

Jamsheed Shorish: And insofar as that is an intersection which you want, you know, is a desirable property of the AMM you're generating, you want something which gives you truthful revelation, but also information aggregation, you could use that entropy representation

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00:52:12.010 --> 00:52:19.050

Jamsheed Shorish: Or perhaps even more abstractly, for other cost functions, generate things that then act as an information

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00:52:19.060 --> 00:52:21.130

Jamsheed Shorish: proxy, like entropy.

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00:52:21.540 --> 00:52:38.109

Jamsheed Shorish: to show the degree to which information is being aggregated when individuals engage in the marketplace. In other words, you would be looking for the reflection in the price space of the underlying disparity of opinion that is occurring in the trading space.

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00:52:38.690 --> 00:52:48.890

Jamsheed Shorish: And that degree to which those two are reflecting each other will tell you whether or not you're getting information aggregation. In other words, is there enough pass-through from what people believe into what you then see?

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00:52:49.200 --> 00:52:53.479

Jamsheed Shorish: As represented by a statistic, like the entropy.

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00:52:54.950 --> 00:53:10.020

Jamsheed Shorish: So, I feel like you're on, like, you're on the right track, in the sense that I think this is a really nice way to see these two things working together. Probably the next step is to really take the information theory side of things and take that seriously as, okay, so this is telling me something about how

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00:53:10.050 --> 00:53:17.299

Jamsheed Shorish: The individual beliefs in the marketplace are, in fact, being aggregated and represented at the market level.

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00:53:17.960 --> 00:53:34.709

Krzysztof Paruch: So that would just be, like... Yeah, right. I mean, I haven't... I have not studied this yet, but in the paper, in the later sections, I mean, there is kind of, like, this, like, statement that the choice of R affects, two properties, which is the bounded law.

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00:53:34.800 --> 00:53:53.800

Krzysztof Paruch: and the information loss, of, of the system. So I think that this is what you're trying to get to here, investigating, kind of, like, this information loss, and investigating this bonded loss, just to kind of, like, to have the financial aspect covered and the informational aspect covered, in this duality.

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00:53:53.800 --> 00:54:03.709

Jamsheed Shorish: Yes, yes, and yeah, I think that's right. The information loss is a little more limited than, as written in the paper, than what we're looking for, or what you would be looking for, in a sense, which is kind of like the...

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00:54:03.860 --> 00:54:12.010

Jamsheed Shorish: You're sort of looking for the... the frictional cost associated with a market... particular market mechanism on information propagation.

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00:54:12.030 --> 00:54:30.120

Jamsheed Shorish: if you could come up with, like, a magic sort of statistic, or a derived statistic, that you could then say, for this market structure, we get this amount of percolation or penetration of information. For this one, this one, and so on, I think it will leverage the individual information gaps that are present and derived from the simple statistic.

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00:54:30.150 --> 00:54:49.629

Jamsheed Shorish: But you are looking for, ultimately, something that will probably become like that graph that you showed at the beginning, where you're looking at the thick markets to thin markets, and watching as the number of individuals falls, just how much more of the market machinery is necessarily brought to bear to pull information out of what they're doing.

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00:54:49.630 --> 00:54:53.790

Jamsheed Shorish: And you can see that, of course, in things like, Markets for which

321

00:54:53.840 --> 00:55:11.970

Jamsheed Shorish: there is a very strong intangible component to the value of the good being sold, like, like art, like selling for art. You know, these are things where you really have to have lots and lots of experts telling you how much something is worth, because when you look at a painting, like a Jackson Pollock, you go, I don't know, is it worth a lot? It's a lot of paint.

322

00:55:11.970 --> 00:55:22.469

Jamsheed Shorish: I don't know. And the question is, like, how do you actually internalize the information that you are supposed to be conveying of your own preferences along the lines of things that are signals from other individuals?

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00:55:22.490 --> 00:55:31.930

Jamsheed Shorish: That path that's being traced is exactly the path of information as it's being aggregated from different individuals with different sets of prior beliefs.

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00:55:33.020 --> 00:55:40.139

Jamsheed Shorish: And so I think there's... there's more work to be done than just taking that as a raw statistic, but I think that's probably the thin edge of the wedge to jump in.

325

00:55:40.960 --> 00:55:50.120

Krzysztof Paruch: Okay, so you said that the entropy could be, like, a starting point in this? So, like, basically the accuracy that I'm trying to map here on the y-axis

326

00:55:50.740 --> 00:55:59.480

Krzysztof Paruch: And what I try to push up, right? So kind of, like, by market design, I want to use a market scoring rule, which is kind of like my complex function.

327

00:55:59.970 --> 00:56:04.350

Krzysztof Paruch: That is moving, kind of, like, this point here as...

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00:56:04.480 --> 00:56:19.800

Krzysztof Paruch: much as possible to the top, and what it means to be at the top needs to be measured somehow, and you say kind of reflective first guess would be to use entropy to start with, and then see if there are other metrics that you can intuit that would be measuring this.

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00:56:21.850 --> 00:56:40.570

Jamsheed Shorish: Yep, exactly, because you've also got, of course, the whole other side of this, which is the properties of market scoring rules, and how you can bend and twist and use those. You've used the one which is the best off-the-shelf with the logarithmic

scoring rule, but of course there are others with different properties, and so there'd be a trade-off space that you probably could generate with regard to how much can you

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00:56:40.570 --> 00:56:54.179

Jamsheed Shorish: sacrifice in terms of aggregation for ease of use, or for computability, for best, let's call them the difference between biasness versus precision, right? So whether or not you're actually, you know, getting a high accuracy, even though you have a large variance in the economy.

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00:56:54.180 --> 00:57:04.630

Jamsheed Shorish: Versus being very, very, very tightly clustered at low accuracy. Maybe that's actually cheaper for individuals who are engaging in the market, and so on. So maybe a nice trade-off space that can be generated from that.

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00:57:07.560 --> 00:57:14.049

Krzysztof Paruch: Okay, so you're basically, like, if you extend this further, what you're trying to say is, okay,

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00:57:15.080 --> 00:57:26.559

Krzysztof Paruch: For particular use cases or specific, yeah, example, like, types of questions, you would, want to kind of, like, use this process, to define

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00:57:27.090 --> 00:57:39.980

Krzysztof Paruch: like, take this R, and kind of, like, come up with a market scoring rule that would, help us to both have the properties of the market scoring rule, like, because it's kind of coming out of this process.

335

00:57:40.700 --> 00:57:55.029

Krzysztof Paruch: And it would, for this particular use case, help us to have very high accuracy, of information aggregation, which would be kind of like the market designer's objective, right? Like, kind of, like, to find the best market scoring rule

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00:57:55.280 --> 00:57:59.529

Krzysztof Paruch: And with the highest information creation properties.

337

00:58:00.110 --> 00:58:10.230

Jamsheed Shorish: And then to kind of circle back to what we were talking about just at the very, very beginning, this would, at the end, allow you to posit certain types of market structures that can make money.

338

00:58:10.510 --> 00:58:20.640

Jamsheed Shorish: And by this, I mean that, as you know, the Abernathy has the notion that a bid and ask spread exists because of indivisibilities.

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00:58:21.040 --> 00:58:35.220

Jamsheed Shorish: Well, if you start to look at the market structure, the underlying state spaces, and say, okay, we have really, really big indivisibilities in these particular spaces, or we have this massive, massive problem spanning certain outcomes with the existing securities.

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00:58:35.220 --> 00:58:45.200

Jamsheed Shorish: Well, that's going to be able to give you just enough leeway to tweak, given our... given then the associated cost function, and given the information propagation.

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00:58:45.210 --> 00:58:55.499

Jamsheed Shorish: The bid and ask spread so that you don't dissuade people from revealing information just enough that you still are able to make money to cover the frictions of the trades.

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00:58:55.500 --> 00:59:02.499

Jamsheed Shorish: Or the outlying, you know, things that are required to service and maintain the ecosystem. Because ultimately, you do need to have that, and

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00:59:02.500 --> 00:59:25.179

Jamsheed Shorish: you know, all too often we see the AMM solutions that just staple costs and fees onto it as the end product, and say, well, and we're going to charge a fee, without really understanding the, you know, the consequences of that, for and especially for prediction markets, you know, something which is an extremely important thing, where the actual goal is to ensure that the accuracy doesn't get slammed, because you've introduced what you thought might have been an innocuous transaction cost.

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00:59:27.970 --> 00:59:29.830

Krzysztof Paruch: Right,

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00:59:31.530 --> 00:59:38.410

Krzysztof Paruch: Okay, so the process still would be, I mean, like, just to summarize what we've learned today, so,

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00:59:38.640 --> 00:59:50.179

Krzysztof Paruch: We know that we can, like, if we state those conditions that we want to have, we seem to have a process that would, given, like, a choice of an R, would spit out a market scoring rule.

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00:59:50.850 --> 01:00:03.700

Krzysztof Paruch: We agree, you and I, that we are not sure if this work always works, but they kind of claim that this is the process. You can do this, you can arrive at it.

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01:00:04.380 --> 01:00:05.890

Krzysztof Paruch: So, the...

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01:00:06.370 --> 01:00:14.000

Krzysztof Paruch: So we have this kind of, like, this instrument, or this machinery, and now what we can say is, given a particular question or a market,

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01:00:14.520 --> 01:00:21.450

Krzysztof Paruch: information revelation Demand, what we are going to do is we're going to select the best R to arrive at an MSR,

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01:00:22.580 --> 01:00:28.970

Krzysztof Paruch: And we would... based... and we would select this R based on the definition of a metric that,

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01:00:29.140 --> 01:00:35.879

Krzysztof Paruch: gives us the highest... the lowest entropy, I would guess, so the best information aggregation, Robert.

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01:00:36.770 --> 01:00:42.689

Krzysztof Paruch: And if we have this, we can implement this and, allow people to trade them.

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01:00:42.950 --> 01:00:51.260

Krzysztof Paruch: it's the best we can do to kind of, like, aggregate information from this set of traders based on this outcome. Is this kind of... is it? This is what we talked about today?

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01:00:52.390 --> 01:01:14.550

Jamsheed Shorish: I mean, from my side, I don't know to say anything from what Peter or David took away, but that sounds great as a research agenda from my side. That gives you both what you've just said, everything up to that point is existence. And then the last thing about transactions costs and can it sustain an economy that is actually profitable for the creation of the market is more like stability properties. Like, if you tweak the system and introduce a bid and ask spread, break apart that single price.

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01:01:14.550 --> 01:01:20.710

Jamsheed Shorish: do you destroy the equilibrium? Do you destroy accuracy? Is it really fragile, or is it really robust to that kind of a small change?

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01:01:20.710 --> 01:01:26.980

Jamsheed Shorish: That would be, like, the last part of the program. But I do want to make sure that, in the time remaining, that David and Peter have anything.

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01:01:27.000 --> 01:01:28.740

Jamsheed Shorish: They'd like to, to add.

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01:01:37.220 --> 01:01:37.930

Krzysztof Paruch: Why not?

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01:01:40.120 --> 01:01:47.260

Krzysztof Paruch: I was wondering, maybe before, maybe they want two more minutes to think about a question, but I was wondering, kind of, like, how this all,

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01:01:47.940 --> 01:01:51.280

Krzysztof Paruch: relates to my initial ambition, so when we...

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01:01:52.070 --> 01:01:58.830

Krzysztof Paruch: when we... when I started on this journey, kind of,

like, my ambition was to find

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01:01:58.980 --> 01:02:06.360

Krzysztof Paruch: To explore the design space with respect to the dimensionality of the of the events.

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01:02:06.620 --> 01:02:08.340

Krzysztof Paruch: So,

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01:02:08.500 --> 01:02:27.430

Krzysztof Paruch: what we're looking at here is a three-dimensional space that is, exclusive and exhaustive, covering, covering all, all the possible outcomes. But, to some extent, kind of, like, the ultimate goal would be to go towards, kind of, like, continuous space, right? So, even looking at how this behaves

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01:02:28.470 --> 01:02:36.680

Krzysztof Paruch: When we are not, you know, querying outcomes, but, continuous distributions from...

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01:02:37.180 --> 01:02:40.640

Krzysztof Paruch: from market participants. So.

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01:02:42.310 --> 01:02:44.640

Jamsheed Shorish: But that's okay, right?

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01:02:45.200 --> 01:03:05.090

Krzysztof Paruch: what I think... I don't know if I can formulate this sufficiently, but I think, like, the question from discrete to continuous in the event space, this is what I'm mostly interested in, if this is even feasible, because this is a program, right? A programming exercise, yeah, where you would be able to...

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01:03:06.190 --> 01:03:13.440

Krzysztof Paruch: of... to step through those, those derivations, for a discrete outcome. But.

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01:03:13.910 --> 01:03:28.680

Jamsheed Shorish: But are you referring to... because I thought you would... you had been looking at the question of extending, like, for sure binary outcomes to a distribution over outcomes, but that the

underlying space of outcomes is still potentially discrete, it's just you have a probability

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01:03:28.910 --> 01:03:40.310

Jamsheed Shorish: distribution over those outcomes? Or are you actually talking about generalizing to an event space where there are actually two events that, in some measurement, are as close as you want them to be?

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01:03:41.540 --> 01:03:53.360

Krzysztof Paruch: Right, the second, the latter. So it's really about, kind of like, distribution markets, like, those are called distribution markets, where you would be trading parameters of, of probability distributions that might be continuous.

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01:03:53.890 --> 01:04:13.880

Jamsheed Shorish: Well, no, but the former, where the underlying event space has a probability distribution, would also be trading the distributions. You'd be trading on the likelihood of, you know, of Biden versus Trump, for example. That would be, like, the distribution, you'd be given the distributions, and those things can be moving continuously, even though you only have two discrete outcomes that can occur in the event space.

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01:04:13.990 --> 01:04:33.929

Jamsheed Shorish: that's different from having a continuum of possible things that can happen, and a distribution over the continua, where you actually need to be able to say, I want square root of 2 away from the current state to be my particular place, and you need to be able to calculate up to a certain level of tolerance, you know, an expansion of international number, and say, I need the probability distribution over that. So I'm just, like, asking, like.

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01:04:34.040 --> 01:04:38.779

Jamsheed Shorish: Which one is necessary for... for what it is you want to do for your program?

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01:04:41.160 --> 01:04:47.480

Krzysztof Paruch: Yeah, good point. I mean, because... you can...

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01:04:48.700 --> 01:04:53.299

Krzysztof Paruch: You can approximate the one with the other, right, in the first case, so, like, you can... you can allow people to...

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01:04:54.090 --> 01:05:01.700

Jamsheed Shorish: Yeah, carefully you can, that's right, but you can always generate... there's a little bit of scope, right, to mess around with a true continuum.

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01:05:01.720 --> 01:05:19.810

Jamsheed Shorish: Because I could create a probability distribution, you know, over a Sirpinski gasket, and then have, like, you know, a completely bonkers probability distribution that is dense everywhere, but has measure zero. And yet, I would still be trading and betting on it. It would look so convoluted that it could mess people up.

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01:05:19.880 --> 01:05:23.319

Jamsheed Shorish: I don't know exactly whether or not that's something which would, you know.

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01:05:23.430 --> 01:05:28.559

Jamsheed Shorish: Pass scrutiny of a legitimate probability distribution when set over a continual.

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01:05:28.580 --> 01:05:33.860

Jamsheed Shorish: I'm just saying a continuum is a huge number of things that can happen, kind of by definition.

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01:05:33.870 --> 01:05:50.459

Jamsheed Shorish: I feel like people's cognitive overload gets hit pretty hard when it's continuous. Even at the level of things like money, what we usually are looking at things that can be even fractions of a cent. We don't look at things that are, like, 10 to the minus 50th of a cent, for example. So...

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01:05:50.490 --> 01:05:58.070

Jamsheed Shorish: I don't know. Maybe, ultimately, you can... you can stay with discreteness, because it actually keeps things manageable, and you can go as small or as large as you want.

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01:05:58.590 --> 01:06:00.400

Jamsheed Shorish: But still have a continuous distribution.

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01:06:00.710 --> 01:06:04.710

Krzysztof Paruch: Yeah, I will for now, I mean, don't worry about that, kind of, like, yeah.

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01:06:05.540 --> 01:06:12.830

Krzysztof Paruch: at least on paper, I have it somewhere that, hey, how will this behave if we are looking at continuous stuff and not on discrete stuff?

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01:06:12.830 --> 01:06:14.300

Jamsheed Shorish: Gotcha. But... Gotcha.

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01:06:14.640 --> 01:06:17.570

Krzysztof Paruch: But let's cross this bridge once we've made this.

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01:06:18.990 --> 01:06:20.500

Krzysztof Paruch: More formal.

392

01:06:22.170 --> 01:06:22.570

Jamsheed Shorish: Nice.

393

01:06:22.570 --> 01:06:29.110

Krzysztof Paruch: Perfect. Yeah, no, thank you, thank you very much for the, for the, participation today, and...

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01:06:29.330 --> 01:06:31.809

Krzysztof Paruch: sticking with me to the end. It was very helpful, too.

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01:06:32.370 --> 01:06:38.350

Krzysztof Paruch: to understand what I'm actually doing, and I hope this makes sense, in some sense, for,

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01:06:38.490 --> 01:06:41.420

Krzysztof Paruch: Yeah, making progress on, on, on those.

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01:06:41.730 --> 01:06:42.910

Krzysztof Paruch: definitions.

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01:06:44.340 --> 01:06:59.000

Jamsheed Shorish: And again, I don't want to speak for anybody else, but I would say that from, you know, from my perspective, if you can make that tie to traditional financial economics, just with regard to state prices, I think you would build a very concrete underpinning.

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01:06:59.240 --> 01:07:12.109

Jamsheed Shorish: For the entire program, because it really strengthens that step of looking at, for example, the negative entropy, and treating it seriously, and saying, look, this really is reflecting

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01:07:12.130 --> 01:07:26.850

Jamsheed Shorish: any kind of risk preference or any kind of underlying distribution, because we know we can transpose this into a state price framework that we've seen already as successful in traditional finance or financial economics. I think that's like a...

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01:07:26.850 --> 01:07:40.220

Jamsheed Shorish: a major call-out. Whether you want to do that earlier or later kind of depends upon how fast you want to proceed in different directions, because you've got a bunch of them. But I do think that's something at the end of the day, probably for things like submitting or presentations, you'd get a question about it.

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01:07:41.350 --> 01:07:51.050

Krzysztof Paruch: Nice, cool, perfect. I need to run, but thank you so much for staying with me, and let's see in two weeks, or in four weeks, I don't remember, but let's... we'll see what the next step will be.

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01:07:51.680 --> 01:07:53.570

Jamsheed Shorish: Sounds great. Alright, thanks guys.

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01:07:53.570 --> 01:07:54.190

David Sisson: Thank you.

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01:07:54.190 --> 01:07:55.510

peterhacker: Thank you. Yeah, thanks, Greg.

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01:07:56.600 --> 01:07:57.190

peterhacker: Bye.

